

Bargaining over Prices

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Seller & buyer bargain price in any trade:

Assume surplus-sharing solution to bargaining problem b/w buyer & seller.

- buyer gets fraction β of surplus
- seller gets fraction $1-\beta$ of surplus
- $\beta \in (0,1)$: bargaining power of buyer

- Diamond (1982)

- If buyer & seller are risk neutral ($\varepsilon \rightarrow \infty$) $\rightarrow$ equivalent to Nash bargaining

Surplus going to seller if price is p_i

$$S_i = \frac{\overset{\text{price in transaction } i}{p_i}}{\underset{\text{aggregate price}}{p}} \cdot \frac{1}{1+x} \cdot \frac{\varepsilon-1}{\varepsilon} \cdot \left(\frac{\mu}{p}\right)^{-1/\varepsilon}$$

$$B_i = \frac{1}{1+x} \cdot \frac{\varepsilon-1}{\varepsilon} \cdot \left(\underline{x_c}^{-1/\varepsilon} - \frac{p_i}{p} \left(\frac{\mu}{p}\right)^{-1/\varepsilon} \right)$$

$$T_i = B_i + S_i = \frac{1}{1+x} \cdot \frac{\varepsilon-1}{\varepsilon} \cdot \underline{x_c}^{-1/\varepsilon}$$

Household's FOC in maximization problem:

$$X c^{-1/2} = [1 + \tau(x)] \left(\frac{N}{P} \right)^{-1/2}$$

Then: $B_i = \frac{1}{1+x} \cdot \frac{1}{2} \cdot \left(\frac{N}{P} \right)^{-1/2} \left(1 + \tau(x) - \frac{P_i'}{P} \right)$

$$T_i' = \frac{1}{1+x} \cdot \frac{1}{2} \cdot [1 + \tau(x)] \left(\frac{N}{P} \right)^{-1/2}$$

$$S_i = \frac{1}{1+x} \cdot \frac{1}{2} \cdot \left(\frac{N}{P} \right)^{-1/2} \cdot \frac{P_i'}{P}$$

Surplus sharing : $\begin{cases} B_i = \beta \cdot T_i \\ S_i = (1-\beta) T_i \end{cases}$

$$\frac{S_i}{T_i'} = 1 - \beta \Rightarrow \frac{P_i'/P}{1 + \tau(x)} = 1 - \beta$$

$$P_i = (1-\beta)(1 + \tau(x)) P$$

P_i is surplus-sharing price in trade i

If $\beta = 0$: $P_i = [1 + \tau(x)] \cdot P$

$\beta = 1$: $P_i' = 0$